Locality and unitarity emerge from hidden zeros in $\mathrm{tr}(\phi^3)$: a by-hand proof at four, five, and six points

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Abstract

We prove that the cyclic hidden one-zeros of color-ordered $\mathrm{tr}(\phi^3)$ tree amplitudes force every non-local coefficient of the general rational ansatz to vanish, and force every triangulation coefficient to take a single common value, at $n = 4, 5, 6$ external legs. The proof uses one move applied at each cyclic zone: *send a chosen subset of free chords to infinity and read off coefficients of the surviving rational function in the remaining variables.* No Laurent expansions are required. Locality (the survivors are exactly triangulations) and unitarity (they share a common scalar) emerge together.

1 Setup

Label the legs of an n-point amplitude cyclically by $1, \ldots, n$ and let $X_{ij} = (p_i + \cdots + p_{j-1})^2$ for $1 \le i < j \le n$, $j - i \ge 2$, $(i, j) \ne (1, n)$. There are $d = n(n-3)/2$ such planar Mandelstam chords. A *triangulation* is a maximal non-crossing set of $n - 3$ chords, and the tree amplitude is $A_n = \sum_T \prod_{c \in T} 1/X_c$ summed over triangulations. We consider the most general rational ansatz with mass dimension $-2(n-3)$ and only planar simple poles,

$$B \;=\; \sum_w \frac{a_w}{\prod_{c \in w} X_c}, \tag{1}$$

where w runs over size-$(n-3)$ multisets of chord indices. A multiset is *local* if its chords form a triangulation.

Non-planar invariants $c_{a,b} = 2p_a \cdot p_b$ satisfy the mesh identity $c_{a,b} = X_{a,b+1} + X_{a+1,b} - X_{a,b} - X_{a+1,b+1}$ (with $X_{a,a+1} = 0$). The hidden one-zero $\mathcal{Z}_r$ is the linear subspace $c_{r,r+2} = \cdots = c_{r,r+n-2} = 0$, $n - 3$ constraints. Solving these via the mesh identity gives the master substitution at $\mathcal{Z}_r$: every chord $X_{r+1,j}$ becomes a $\mathbb{Q}$-linear function of the remaining (*free*) chords, with the special chord $X_* := X_{r,r+2}$ appearing on every right-hand side. We write F_r for the free set excluding X_*. Rodina has shown $A_n|_{\mathcal{Z}_r} \equiv 0$; our goal is the converse: assuming $B|_{\mathcal{Z}_r} = 0$ for all r, show $B = a \cdot A_n$ for some scalar a.

Lemma 1 (The kill move)**.** *Suppose $B|_{\mathcal{Z}_r} = 0$. Let $S \subseteq F_r \cup \{X_*\}$ be any subset of free chords held arbitrary, and let every chord outside S go to ∞. Whenever a non-bare master substitution $X_s = X_f - X_*$ has its companion X_f outside S, also let $X_f \to \infty$ (otherwise $1/X_s = 1/(X_f - X_*)$ would survive as a non-monomial rational factor, blocking coefficient extraction). Then the surviving terms of $B|_{\mathcal{Z}_r}$ form a Laurent polynomial in S, and each coefficient on a distinct monomial vanishes.*

Proof. After the master substitution, $B|_{\mathcal{Z}_r}$ is a rational function in the $|F_r| + 1$ free chords. Sending a chord to infinity drops every monomial of B whose denominator contains it. The companion condition ensures all surviving monomials are Laurent monomials in S; distinct multisets give distinct monomials, so linear independence forces individual vanishing. □

We use Lemma 1 in two flavors:

- **Step 1** drops X_* from S. Surviving multisets are subsets of F_r, and coefficients are forced to zero individually.
- **Step 2** keeps $X_* \in S$ and exploits $X_{\text{bare}} = -X_*$. All non-bare substituted chords get dropped (by sending their companions outside S to infinity). The surviving polynomial in S pairs multisets glued by the sign of $-X_*$, producing equalities of original coefficients.

2 Four points

The square has only X_{13}, X_{24} and the ansatz $B = a_{X_{13}}/X_{13} + a_{X_{24}}/X_{24}$ has *no* non-local multisets: with $n - 3 = 1$, every single chord is itself a triangulation, and there is no room for a crossing. Locality is automatic. The single zero $\mathcal{Z}_1$ gives $X_{24} = -X_{13}$, so $B|_{\mathcal{Z}_1} = (a_{X_{13}} - a_{X_{24}})/X_{13} = 0$, hence $a_{X_{13}} = a_{X_{24}}$. This is unitarity. $B = a \cdot A_4$.

3 Five points

3.1 Setup

The pentagon has 5 chords, all class 1 (short). The ansatz has $\binom{5+1}{2} = 15$ coefficients: 5 tree triples (the fan triangulations), 5 double poles, 5 crossing pairs. We must kill the latter 10 and equate the 5 tree triples. The five zones are

Zone	Master subs		X_*	F_r
$\mathcal{Z}_1$	$X_{24} = X_{14} - X_{13}$	$X_{25} = -X_{13}$	X_{13}	$\{X_{14}, X_{35}\}$
$\mathcal{Z}_2$	$X_{35} = X_{25} - X_{24}$	$X_{13} = -X_{24}$	X_{24}	$\{X_{25}, X_{14}\}$
$\mathcal{Z}_3$	$X_{14} = X_{13} - X_{35}$	$X_{24} = -X_{35}$	X_{35}	$\{X_{13}, X_{25}\}$
$\mathcal{Z}_4$	$X_{25} = X_{24} - X_{14}$	$X_{35} = -X_{14}$	X_{14}	$\{X_{24}, X_{13}\}$
$\mathcal{Z}_5$	$X_{13} = X_{35} - X_{25}$	$X_{14} = -X_{25}$	X_{25}	$\{X_{35}, X_{24}\}$

Each zone has one special X_*, one non-bare substitution $X_s = X_f - X_*$ with companion $X_f \in F_r$, one bare $X_{\text{bare}} = -X_*$, and two free chords.

3.2 Step 1 at $\mathcal{Z}_1$ (three kills)

Send $X_{13} \to \infty$ holding X_{14}, X_{35} fixed: then $X_{24}, X_{25} \to -\infty$ and every B-monomial involving any of X_{13}, X_{24}, X_{25} drops out. Surviving (with $a_{ij,kl} := a_{X_{ij}X_{kl}}$):

$$\sum_{w \subseteq \{X_{14}, X_{35}\}} \frac{a_w}{\prod X_w} = \frac{a_{14,14}}{X_{14}^2} + \frac{a_{14,35}}{X_{14}X_{35}} + \frac{a_{35,35}}{X_{35}^2} = 0,$$

so by independence $a_{14,14} = a_{14,35} = a_{35,35} = 0$.

3.3 Step 2 at $\mathcal{Z}_1$ (pair equalities)

To use the bare relation $X_{25} = -X_{13}$ cleanly we must drop the non-bare substituted chord $X_{24} = X_{14} - X_{13}$. Sending $X_{14} \to \infty$ does this: then $X_{24} \to \infty$ as well, and *both* X_{14}- and X_{24}-involving monomials drop from the surviving sum. This is essential — if we kept X_{14} finite while letting X_{24} contribute, every X_{24}-monomial would carry a factor $1/X_{24} = 1/(X_{14} - X_{13})$, which is not a Laurent monomial in $\{X_{13}, X_{14}\}$ and would block clean coefficient extraction.

Holding X_{13}, X_{35} fixed and sending $X_{14} \to \infty$, the survivors are multisets in $\{X_{13}, X_{25}, X_{35}\}$, glued by $X_{25} = -X_{13}$. Below we abbreviate $a_{X_{ij}X_{kl}}$ as $a_{ij,kl}$:

$$\frac{a_{13,13}}{X_{13}^2} - \frac{a_{13,25}}{X_{13}^2} + \frac{a_{25,25}}{X_{13}^2} + \frac{a_{13,35}}{X_{13}X_{35}} - \frac{a_{25,35}}{X_{13}X_{35}} + \frac{a_{35,35}}{X_{35}^2} = 0.$$

Collecting on $1/X_{13}^2,\ 1/(X_{13}X_{35}),\ 1/X_{35}^2$:

$$\boxed{a_{13,13} - a_{13,25} + a_{25,25} = 0, \qquad a_{13,35} = a_{25,35}, \qquad a_{35,35} = 0 \text{ (already from Step 1).}}$$

The first identity, combined with Step 1 kills of $a_{13,13}$ (via $\mathcal{Z}_3, \mathcal{Z}_4$) and $a_{25,25}$ (via $\mathcal{Z}_2, \mathcal{Z}_3$), forces the crossing pair $a_{13,25} = 0$ — a second route to the same kill. The middle identity equates two fan triangulations sharing X_{35}.

3.4 Cycling and the union

Cycling $1 \to 2 \to \cdots \to 5$ rotates Step 1 and Step 2 across all five zones. Below we abbreviate $a_{X_{ij}X_{kl}}$ as $a_{ij,kl}$:

Zone	Step 1 kills	Step 2 unitarity equality
$\mathcal{Z}_1$	$a_{14,14},\ a_{35,35},\ a_{14,35}$	$a_{13,35} = a_{25,35}$
$\mathcal{Z}_2$	$a_{25,25},\ a_{14,14},\ a_{14,25}$	$a_{14,24} = a_{13,14}$
$\mathcal{Z}_3$	$a_{13,13},\ a_{25,25},\ a_{13,25}$	$a_{25,35} = a_{24,25}$
$\mathcal{Z}_4$	$a_{24,24},\ a_{13,13},\ a_{13,24}$	$a_{13,14} = a_{13,35}$
$\mathcal{Z}_5$	$a_{35,35},\ a_{24,24},\ a_{24,35}$	$a_{24,25} = a_{14,24}$

Locality. The Step 1 union covers the 5 double poles ($a_{ij,ij}$, each in two zones) and the 5 crossing pairs ($a_{14,35}$ at $\mathcal{Z}_1$, $a_{14,25}$ at $\mathcal{Z}_2$, $a_{13,25}$ at $\mathcal{Z}_3$, $a_{13,24}$ at $\mathcal{Z}_4$, $a_{24,35}$ at $\mathcal{Z}_5$). All ten non-locals are killed.

Unitarity. The five Step 2 equalities form a 5-cycle on the five fan triangulations: $a_{13,35} \overset{\mathcal{Z}_1}{=} a_{25,35} \overset{\mathcal{Z}_3}{=} a_{24,25} \overset{\mathcal{Z}_5}{=} a_{14,24} \overset{\mathcal{Z}_2}{=} a_{13,14} \overset{\mathcal{Z}_4}{=} a_{13,35}$. All five fan coefficients are equal to a common a, hence $B = a \cdot A_5$.

Theorem 1 ($n = 5$)**.** *The five hidden one-zeros of the pentagon force $B = a \cdot A_5$ for a unique scalar a. Locality and unitarity emerge simultaneously from the kill-and-equate enumeration.*

4 Six points

4.1 Setup

The hexagon has 9 chords: six short ($X_{13}, X_{24}, X_{35}, X_{46}, X_{15}, X_{26}$, each enclosing one vertex) and three long (X_{14}, X_{25}, X_{36}, each enclosing two). The ansatz has $\binom{9+2}{3} = 165$ coefficients, of which 14 are tree triples ($C_4 = 14$ triangulations) and 151 are non-local. Each $\mathcal{Z}_r$ has three master substitutions, two non-bare and one bare. For $\mathcal{Z}_1$:

$$X_{24} = X_{14} - X_{13}, \qquad X_{25} = X_{15} - X_{13}, \qquad X_{26} = -X_{13}, \qquad X_* = X_{13}.$$

The free set is $F_1 = \{X_{35}, X_{46}, X_{15}, X_{14}, X_{36}\}$ (3 short, 2 long). The other zones are cyclic rotations.

4.2 Step 1 at $\mathcal{Z}_1$ (35 kills)

Send $X_{13} \to \infty$, F_1 fixed. Then $X_{24}, X_{25}, X_{26} \to \pm\infty$, and the surviving terms are exactly the multisets in F_1. Writing $a_{abc} := a_{X_a X_b X_c}$:

$$\sum_{(a,b,c)\subseteq F_1} \frac{a_{abc}}{X_a X_b X_c} = 0 \quad \Longrightarrow \quad \boxed{a_{abc} = 0 \text{ for every multiset of size 3 from } F_1.}$$

There are $\binom{5+2}{3} = 35$ such multisets.

4.3 Step 2 at $\mathcal{Z}_1$ (equalities and 3-term identities)

To use the bare relation $X_{26} = -X_{13}$ cleanly we must drop *both* non-bare substitutions $X_{24} = X_{14} - X_{13}$ and $X_{25} = X_{15} - X_{13}$. If either survived in the sum, its $1/X_{24} = 1/(X_{14} - X_{13})$ or $1/X_{25} = 1/(X_{15} - X_{13})$ factor would not be a Laurent monomial in our chosen variables, blocking clean coefficient extraction. Sending both companions $X_{14}, X_{15} \to \infty$ achieves this: $X_{24}, X_{25} \to \infty$ also, and every B-monomial involving any of $X_{14}, X_{24}, X_{15}, X_{25}$ drops.

Keep $S = \{X_{13}, X_{35}, X_{46}, X_{36}\}$ arbitrary. Surviving multisets draw from $\{X_{13}, X_{26}, X_{35}, X_{46}, X_{36}\}$, glued by $X_{26} = -X_{13}$. Each surviving B-monomial becomes a Laurent monomial in S, with sign $(-1)^k$ where k counts factors of X_{26}. Writing $a_{ab,c} := a_{X_a X_b X_c}$:

$$\boxed{a_{13^3} - a_{13^2,26} + a_{13,26^2} - a_{26^3} = 0, \quad a_{13^2,b} - a_{13,26,b} + a_{26^2,b} = 0, \quad a_{13,a,b} = a_{26,a,b}}$$

for $X_a, X_b \in \{X_{35}, X_{46}, X_{36}\}$. The third identity is a direct equality of two coefficients; whether each is between two triangulations or two non-locals depends on $\{X_a, X_b\}$ and is sorted out by the orbit accounting below. Cyclic rotations to $\mathcal{Z}_2, \ldots, \mathcal{Z}_6$ give the full Step 2 family.

4.4 Cycling: locality and unitarity

Locality at $n = 6$. Step 1 across all six zones kills every multiset whose support sits inside some F_r (129 of the 151 non-local multisets, after accounting for overlap). The remaining 22 non-local multisets touch every vertex; they fall into five cyclic-symmetry orbits:

$$\text{(a) } X_{r,r+2}^2 X_{r+3,r+5}, \text{ (b) } \{X_{r,r+2}, X_{r+1,r+3}, X_{r+3,r+5}\}, \text{ (c) } \{X_{r,r+2}, X_{r+1,r+3}, X_{r+4,r+6}\},$$
$$\text{(d) } \{X_{r,r+2}, X_{r+3,r+5}, X_{r,r+3}\}, \text{ (e) } \{X_{14}, X_{25}, X_{36}\}.$$

For each orbit, two cyclically-shifted Step 2 identities (case A at $\mathcal{Z}_r$, case B at $\mathcal{Z}_{r-1}$) give both an equality $a_M = a_{M'}$ *and* a difference $a_M - a_{M'} = 0$, forcing $a_M = 0$. The detailed orbit verification is in the companion note (#17 page 4 et seq.); each verification is one application of Step 2, so all 22 remaining non-locals die, completing locality.

Unitarity at $n = 6$. The Step 2 identities $a_{13,a,b} = a_{26,a,b}$ equate adjacent triangulation coefficients; each pair (X_{13}, X_{26}) corresponds to a flip in the hexagon's associahedron K_5. Cyclically, every flip in the 14-triangulation flip-graph is realized by some Step 2 equality, the flip-graph is connected, so all 14 tree-triple coefficients equal a common a, and $B = a \cdot A_6$.

Theorem 2 ($n = 6$)**.** *The six hidden one-zeros of the hexagon force $B = a \cdot A_6$. All* 151 *non-local coefficients vanish (*129 *by Step 1,* 22 *by cyclic-shifted Step 2 on five orbits) and the* 14 *triangulation coefficients collapse to a single common value via Step 2 flips.*

General-n outlook. The two-step move applies at every n. Open: (i) does cyclic-shifted Step 2 cover all multisets outside any F_r? (verified $n \leq 6$, numerical $n = 7$); (ii) does Step 2 cover all flips of K_{n-1}? (verified $n \leq 6$). Closing both is the central goal of the ongoing program.